# A Tutorial on Graph RAG & Meta KDD Cup

2026.1.30

Data Systems Lab

# Overview

- Introduction
  - Retrieval-augmented Generation (RAG)
  - Graph RAG
- Meta KDD Cup
  - Benchmark
  - Baseline Approach

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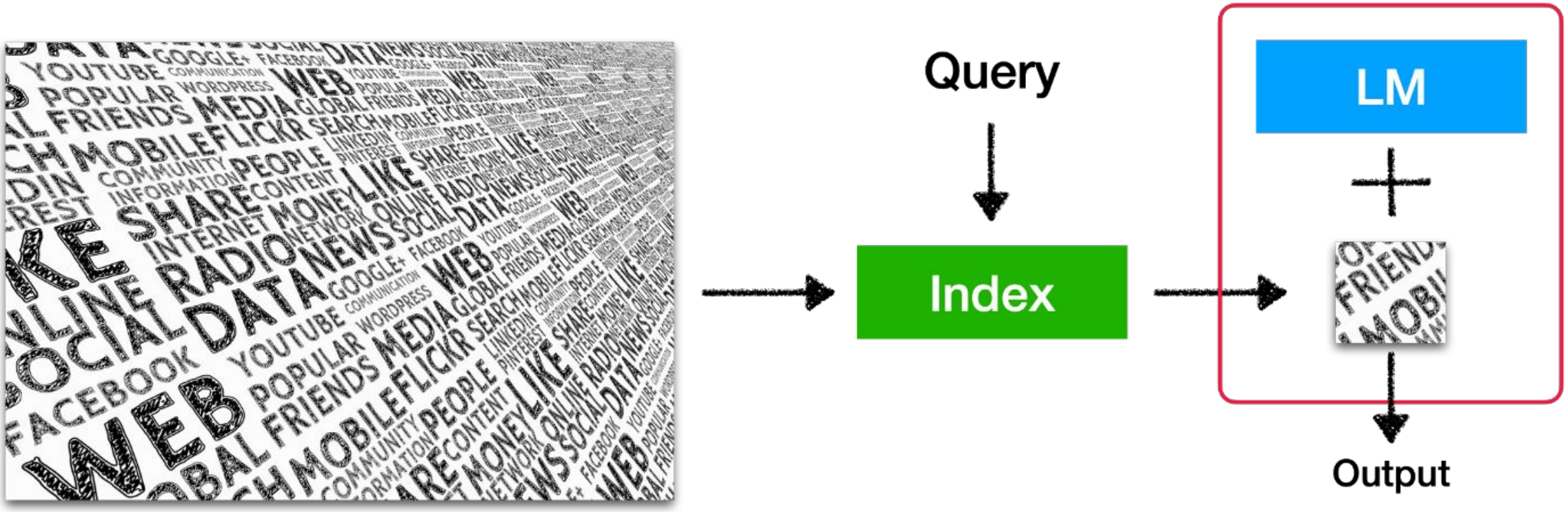
# Retrieval-augmented Generation (RAG)




Datastore

Incorporate Datastore at inference

Slides adapted from the ACL-23 tutorial:
Retrieval-based Language Models and Applications

# RAG Mitigates Weakness of Large Language Models

**Hallucinations**

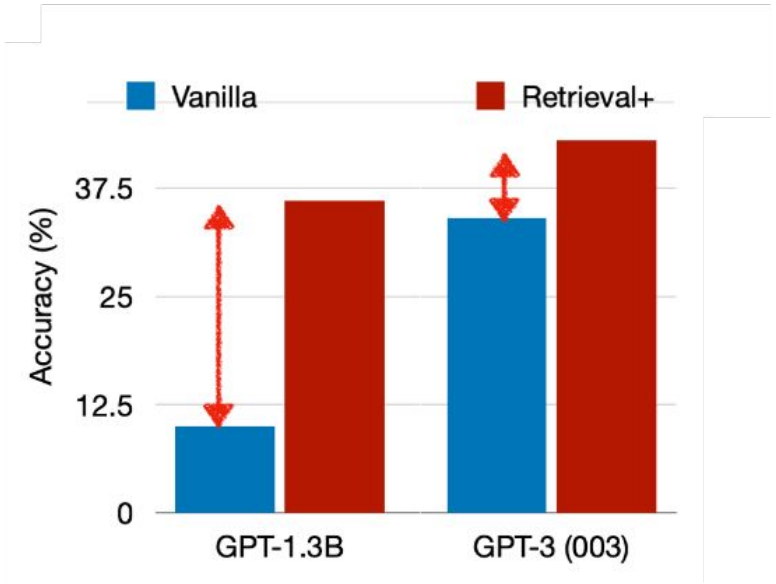
Costs of adaptations

Copyright / privacy

Large parameter size

Significant improvements across model scale, with larger gain with smaller LM

QA




Mallen*, Asai* et al., When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories (ACL, Best Video; Oral) 2023.

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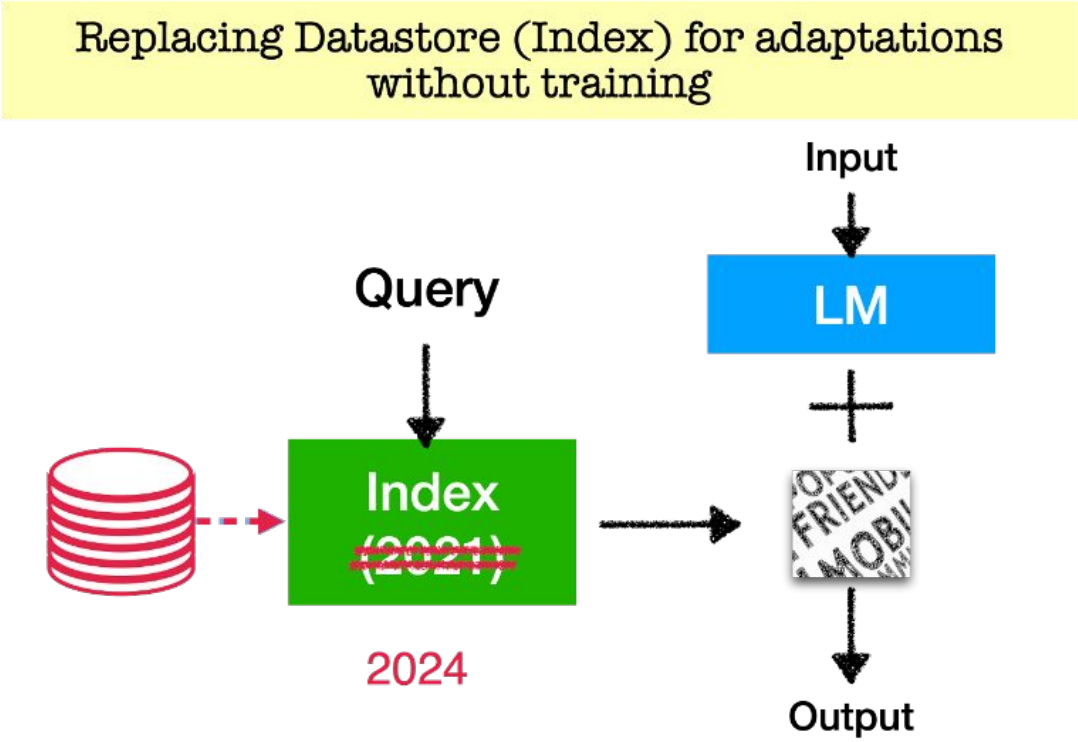


Izacard*, Lewis* et al., Atlas: Few-shot Learning with Retrieval Augmented Language Models. JMLR 2023.

# RAG Mitigates Weakness of Large Language Models

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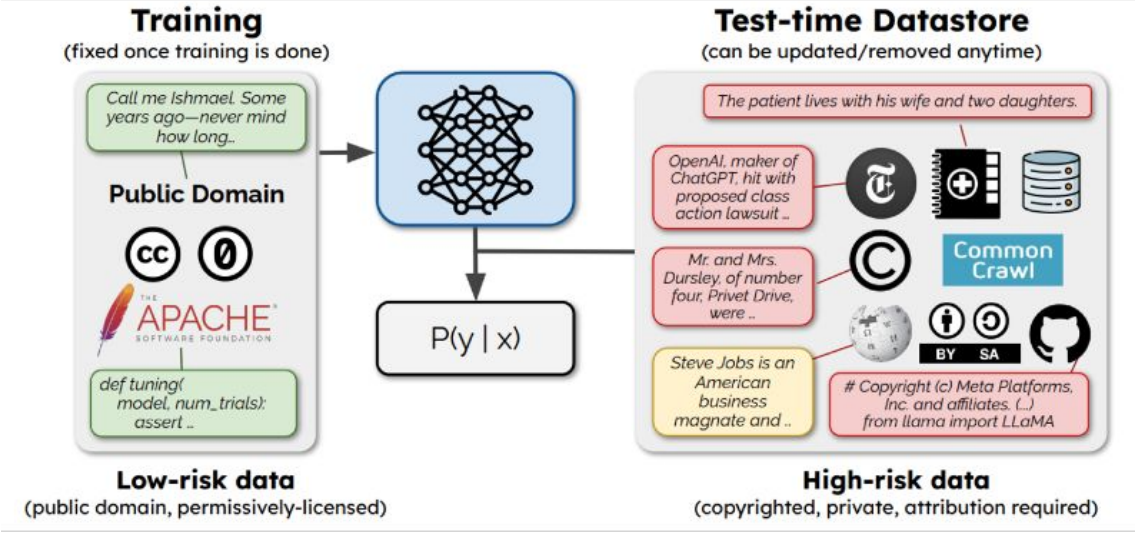
Segregating copyright-sensitive data from pre-training data




Min* and Gururangan* et al., SILO Language Models: Isolating Legal Risk In a Nonparametric Datastore. ICLR 2024

# RAG Mitigates Weakness of Large Language Models

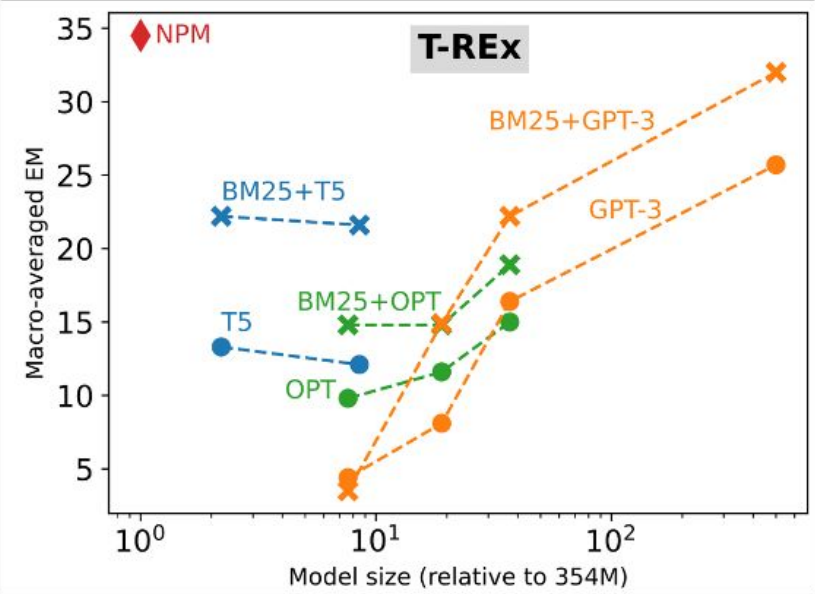
Hallucinations

Costs of adaptations

Copyright / privacy

**Large parameter size**

Models with much less parameters can outperforms much larger models!




Min et al., Nonparametric Masked Language Modeling. Findings of ACL 2023.

# One of the Hottest Topics in VLDB 2024: (Graph)RAG

## Vector Databases

A2 Panel Vector Databases: What's Really New and What's Next?

A1 Data management and support for ML/AI

**Experimental Analysis of Large-scale Learnable Vector Storage Compression**
*University)*; Penghao Zhao (Peking University); Xupeng Miao (Carnegie Mellon Uni (Peking University); Bin Cui (Peking University)*

**SingleStore-V: An Integrated Vector Database System in SingleStore**
*Zhang (Purdue University - West Lafayette); Sasha Podolsky (SingleStore); Zhou Sun (SingleStore); Robert Walzer (SingleStore); Jianguo Wang (Purdue*

**Chat2Data: An Interactive Data Analysis System with RAG, Vector Databases and LLMs** *xi*
*Guoliang Li (Tsinghua University)**

## Graph Databases

NebulaGraph

Industry Talk: Integrating GenAI with Graph: Innovations and Insights from NebulaGraph

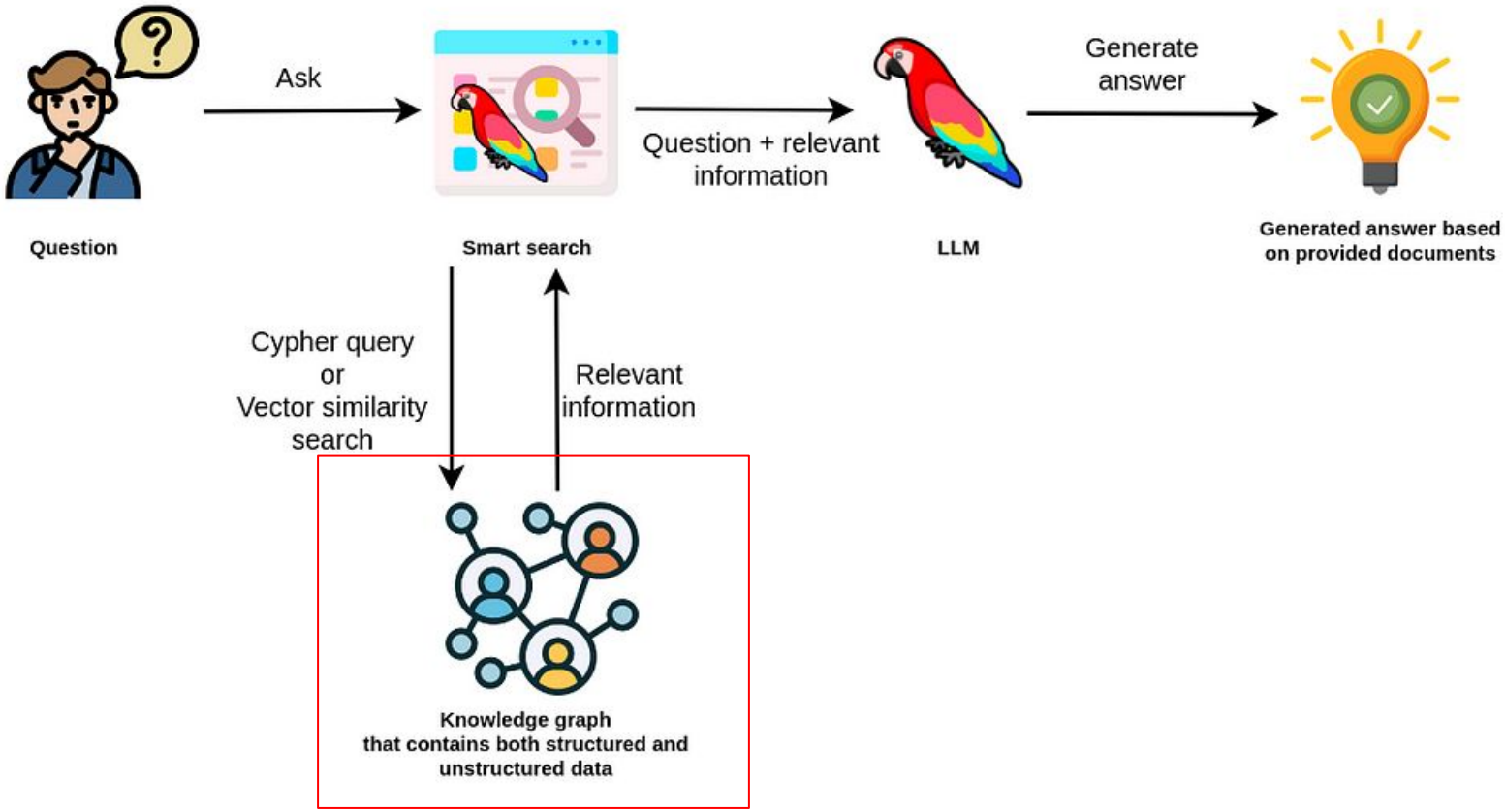
Siwei Gu & Yihang Yu (NebulaGraph, China)

LLM+KG Workshop

neo4j

Microsoft

# GraphRAG Overview




**GraphRAG Overview (From Neo4j Document)**

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# Meta KDD Cup 2024

- 2300+ participants, 384 teams, 5600+ submissions
- CRAG benchmark was listed in HuggingFace "Daily papers". Hugging Face
- Our team was the only Korean team to receive an award, achieving First Place in the Comparison Question category for Tasks 1, 2, and 3!!

Comprehensive RAG Benchmark Aims t...
analyticsindiamag.com

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LAST UPDATED JUNE 13, 2024 • IN AI NEWS & UPDATE

## Comprehensive RAG Benchmark Aims to Advance Retrieval-Augmented Question Answering

Meta AI Introduces New Dataset and Evaluation to Push Limits of Knowledge-Grounded Language Models

ADaSci

Members Area

Deep Dives

## Enhancing Retrieval-Augmented Generation in NLP with CRAG

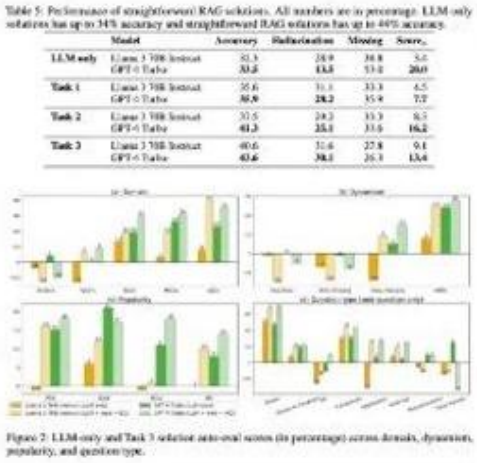
Learn how CRAG benchmarks Retrieval-Augmented Generation (RAG) systems for reliable and creative question-answering in NLP.

By Shreepradha Hegde Published on June 11, 2024

Home > Technology > AI Shorts > Advancing Reliable Question Answering with the CRAG Benchmark

## Advancing Reliable Question Answering with the CRAG Benchmark

By Mohammad Asjad - June 11, 2024



https://arxiv.org/abs/2406.04744

0 SHARES

Large Language Models (LLMs) have revolutionized Natural Language Processing (NLP), particularly in Question Answering (QA). However, hallucination

marktechpost.com

Meta

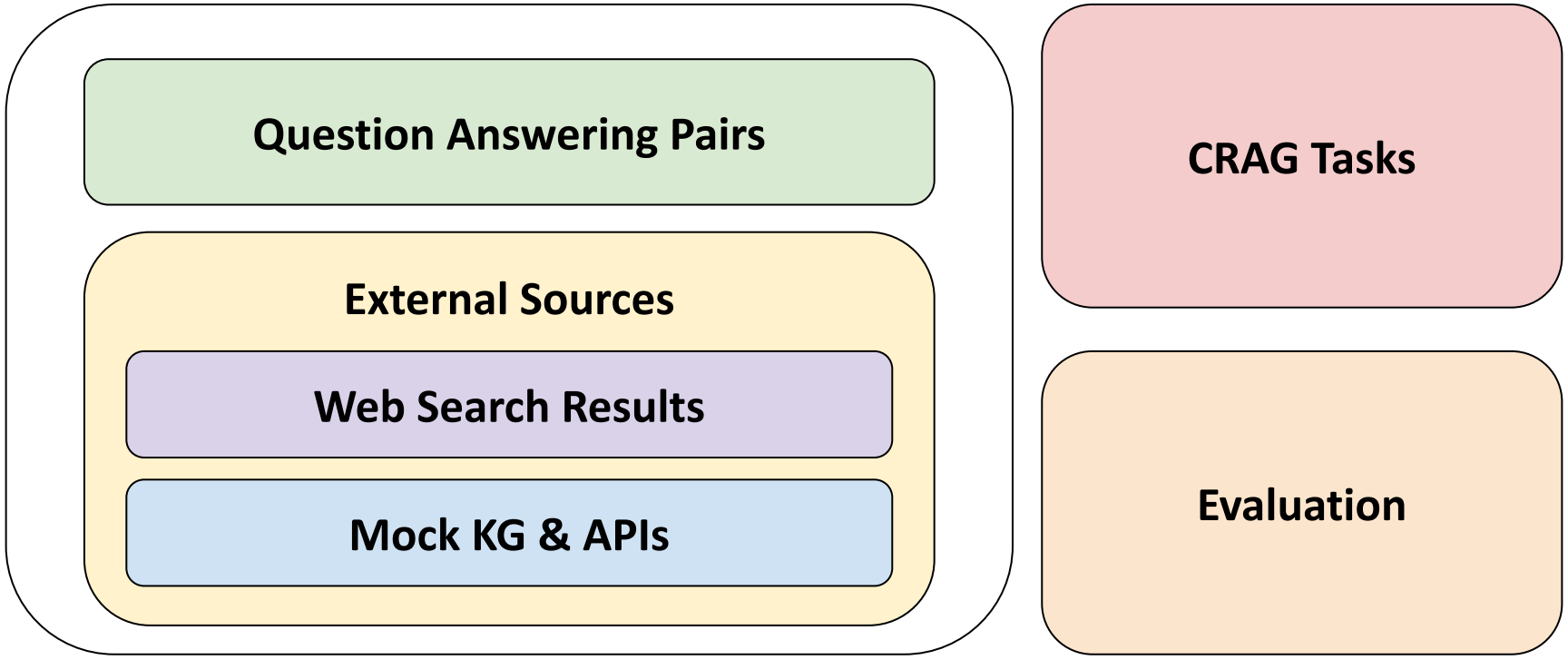
## Advancing Reliable Question Answering with the CRAG Benchmark

TLDR A benchmark called CRAG has been proposed to improve reliable question answering systems by addressing hallucination issues in Large Language Models. The benchmark contains diverse QA pairs, covers varying entity popularity and temporal spans, and offers three tasks to evaluate RAG solutions. Evaluations show that incorporating Show more

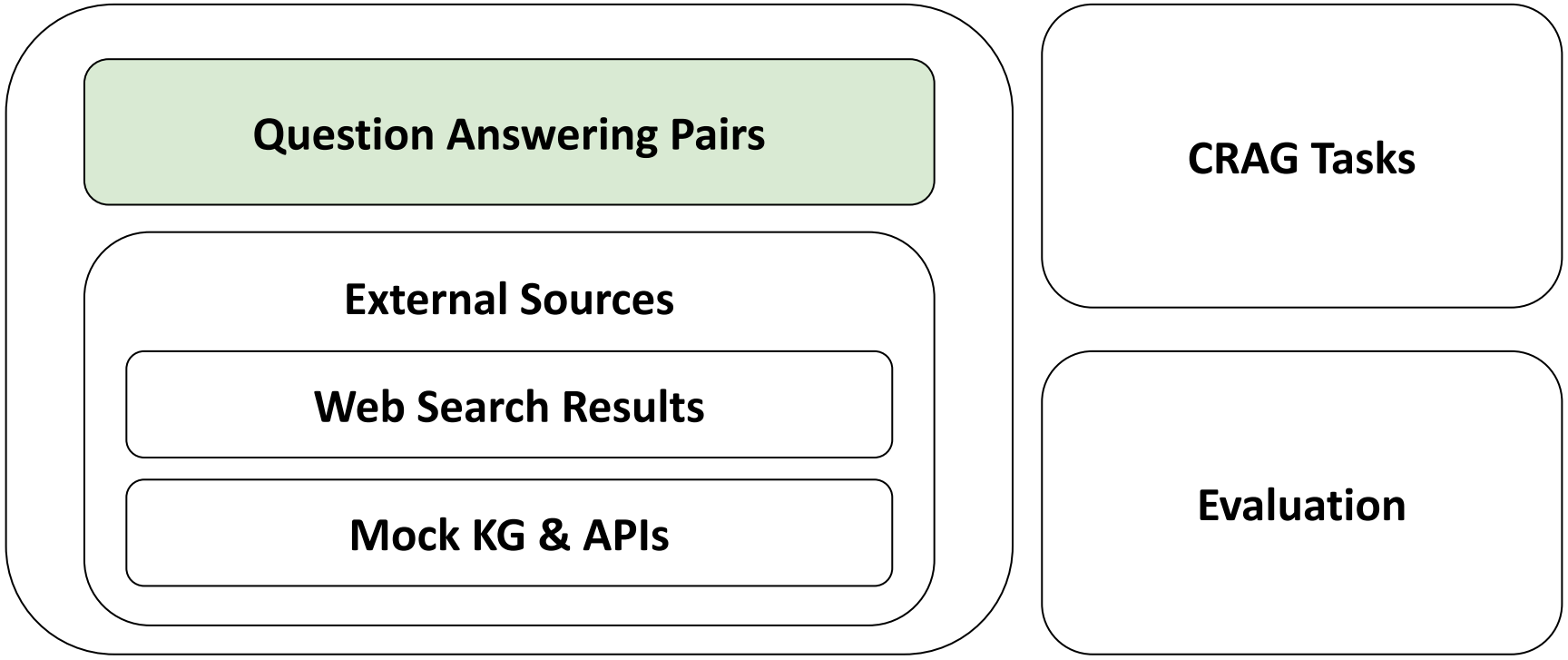
#nlp #rag

Jun 11 • 3m read time

# CRAG Benchmark Overview

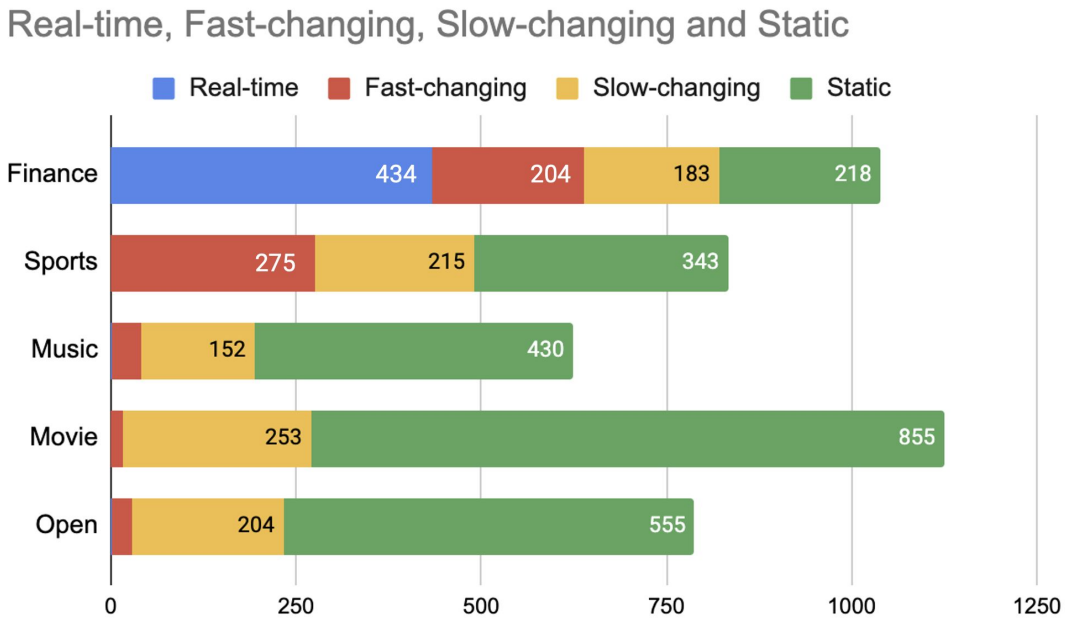


# CRAG Benchmark Overview



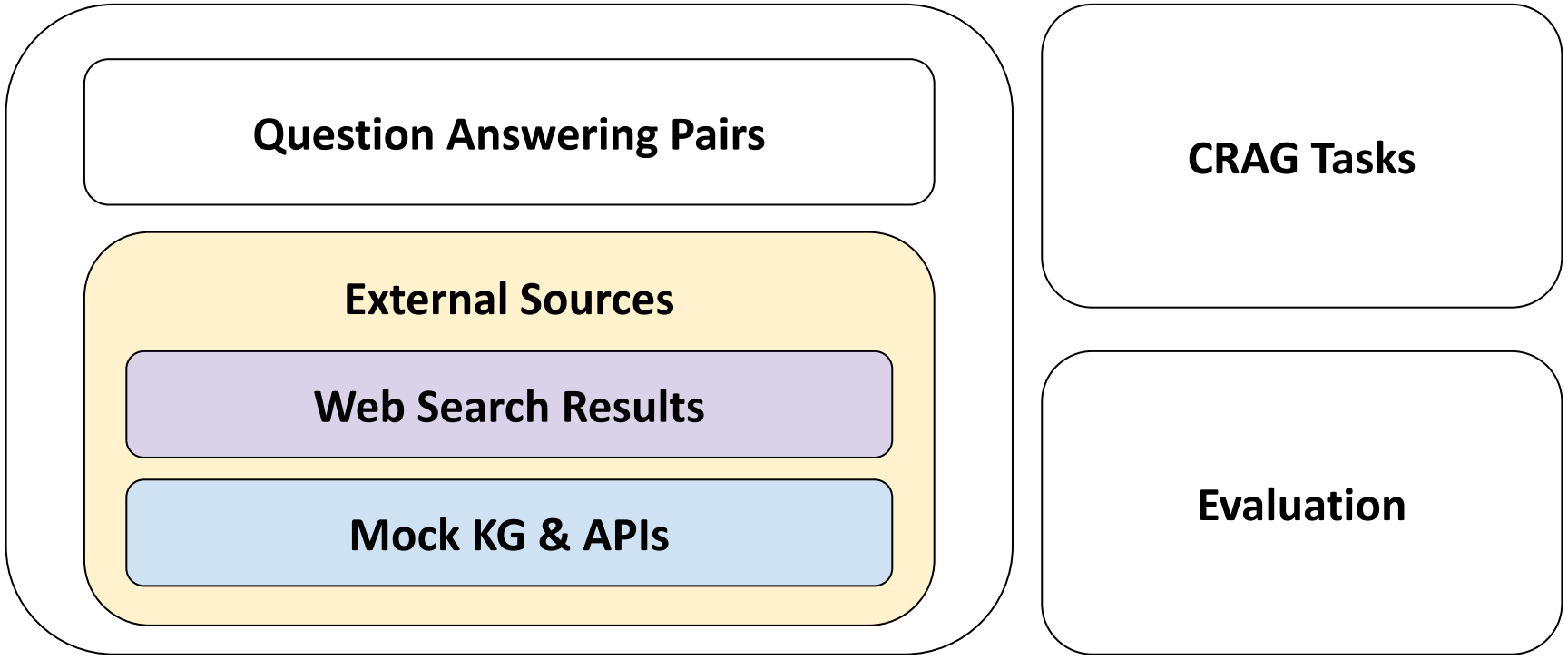
# Question Answering Pairs

- 4400+ QA pairs from 5 domains (Finance, Sports, Music, Movie, Encyclopedia)
- Questions for static, slow-changing, fast-changing, and real-time information
- Questions for head, torso, and tail entities
- Simple-fact questions and complex questions




| Total | Simple | Simple w. Cond | Set | Comparison | Aggregation | Multi-hop | Post-Processing | False Premise |
|---|---|---|---|---|---|---|---|---|
| 4409 | 1205 | 689 | 403 | 546 | 489 | 382 | 180 | 525 |

# CRAG Benchmark Overview

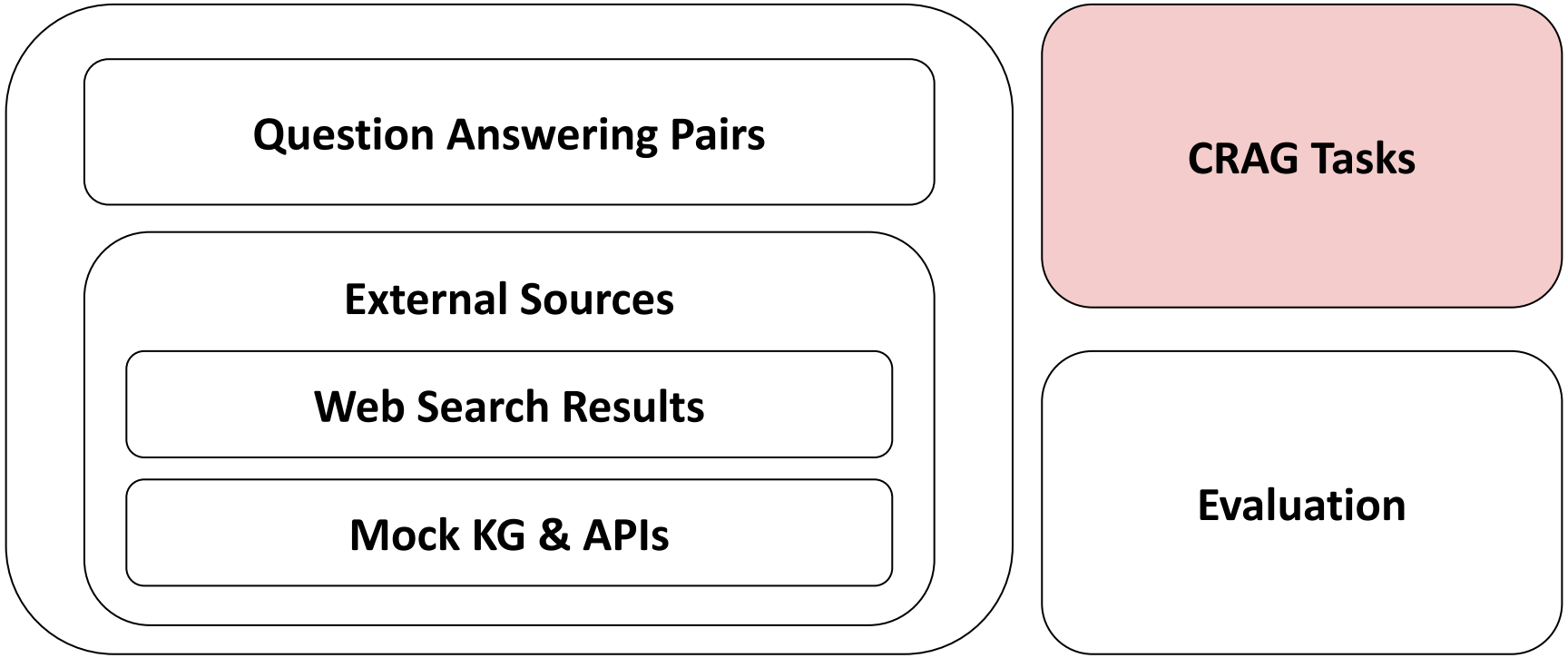


# External Sources

- Web Search Results: 50 webpages for each question from BraveAPI web search
- Mock KG & APIs
  - Mock KG: 2.6M entities
  - Mock APIs: 38 mock APIs

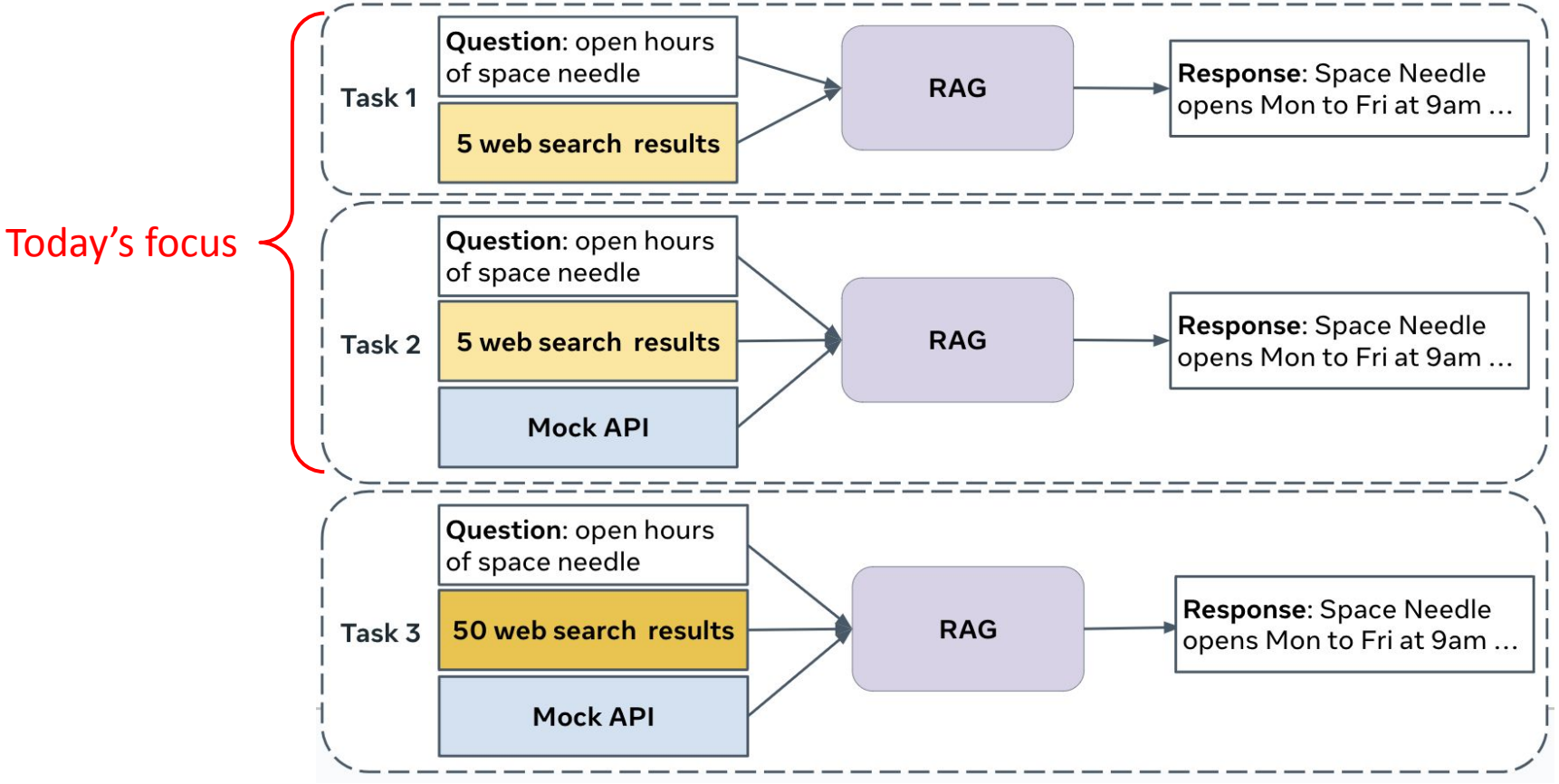
| | |
|---|---|
| **Q1:**<br>What's the latest film that walt becker has directed?<br>**Q2:**<br>Which one of these came out earlier, the greater meaning of water or small town ecstasy? | **API for Q1:**<br>get_movie_person_crew(None,"walt becker", eq(job, "Director"));<br>sort(None,-year)["movie_name"]<br>**API for Q2:**<br>get_movie("greater meaning of water")["release_date"];<br>get_movie("small town ecstasy")["release_date"] |

# CRAG Benchmark Overview



# CRAG Tasks

Three tasks build up information gradually to test different capabilities of RAG systems.



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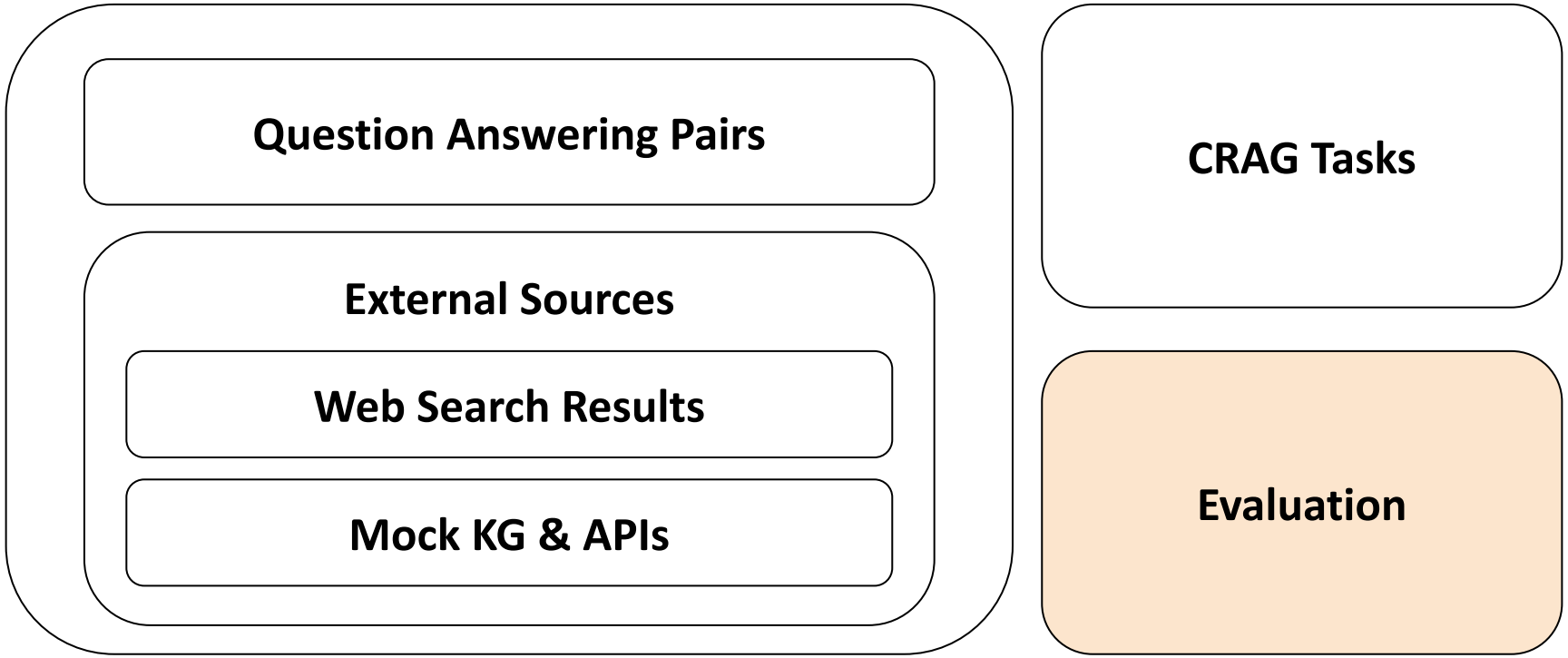
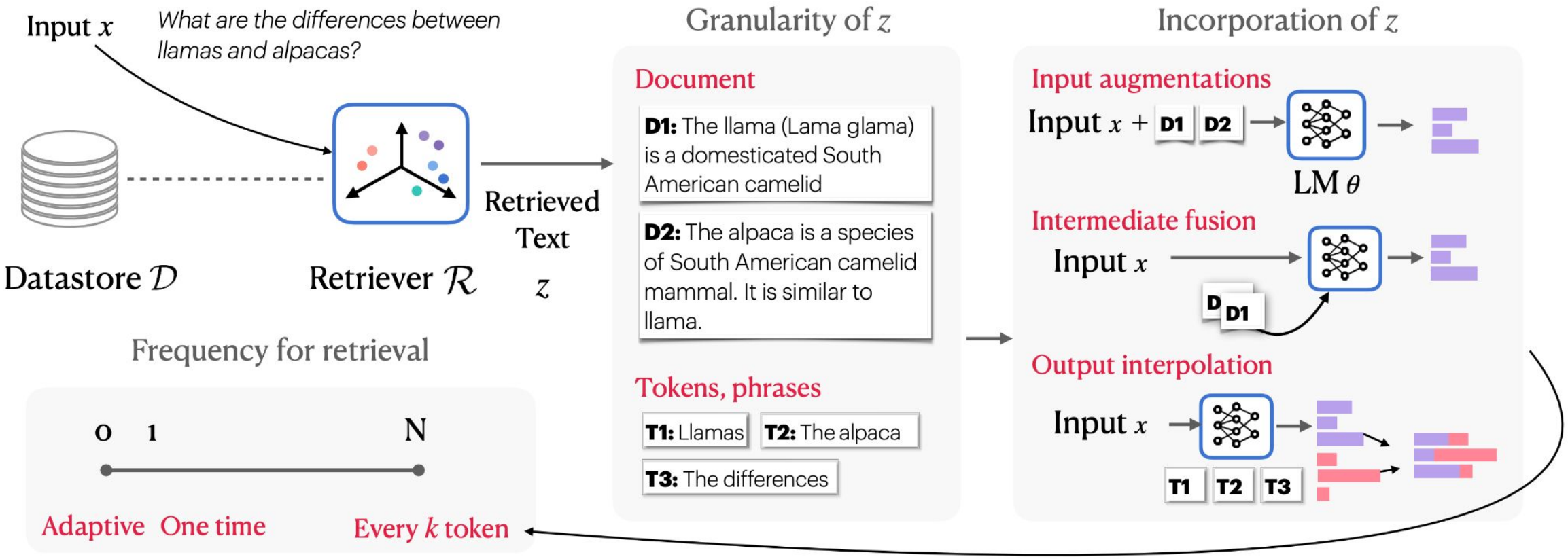


# Evaluation

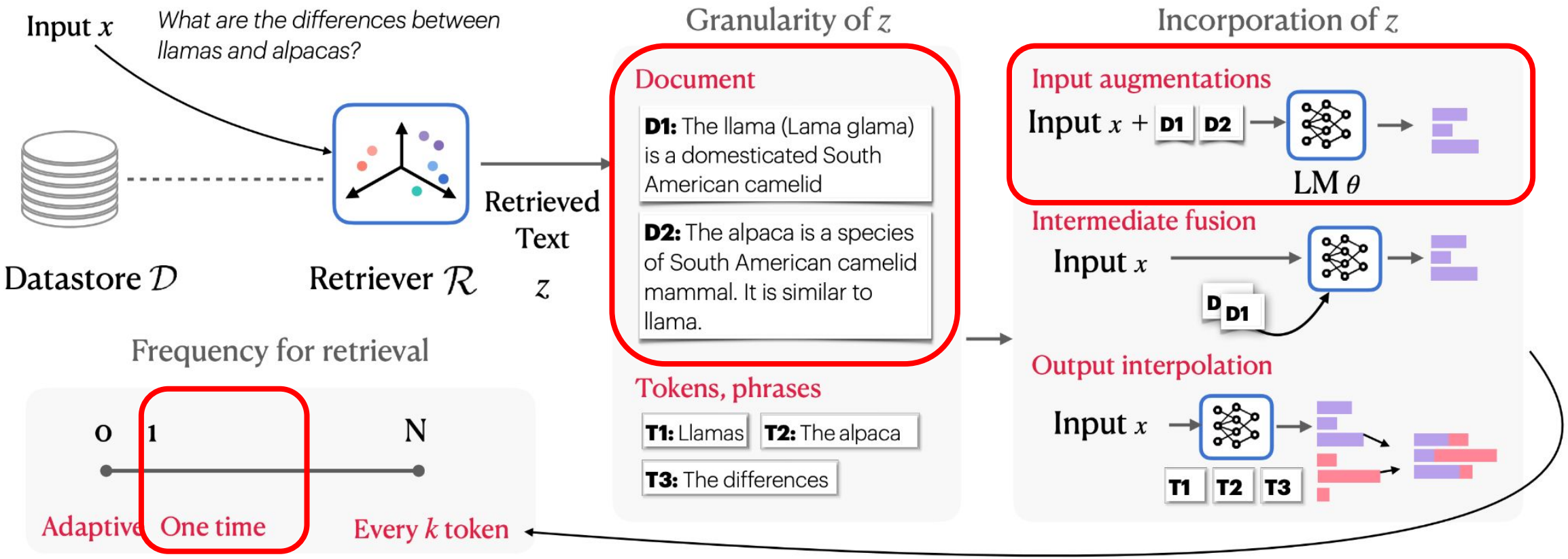
- Metrics: CRAG Score = Exact Accuracy + 0.5 * Accuracy - Hallucination rate
  - **Exact Accuracy:** The percentage of questions for which the generated answer exactly matches the ground truth answer.
  - **Accuracy:** The percentage of questions for which the generated answer is not exact but has the same meaning as the ground truth.
  - **Hallucination:** The percentage of questions for which an incorrect answer was generated.
  - **Missing:** The percentage of questions where the response was "I don’t know."
- Evaluation
  - Auto-eval (with GPT-4)
  - Manual-eval (with Human)

# Diverse Architectures of RAG




Asai, Akari, et al. "Reliable, adaptable, and attributable language models with retrieval." arXiv preprint arXiv:2403.03187 (2024).

# Baseline Approach

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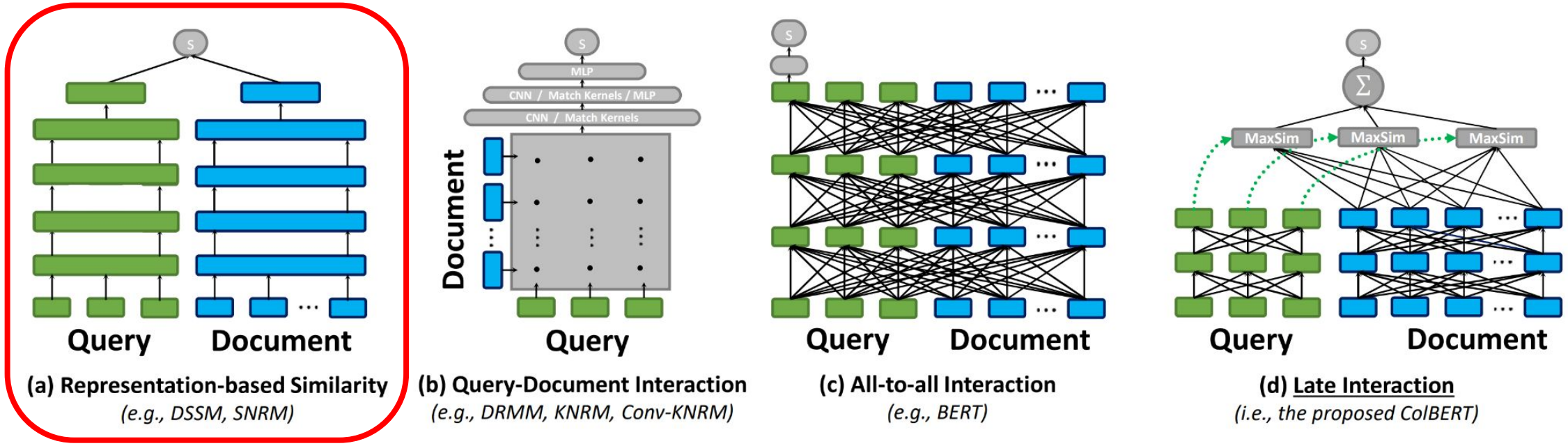
# Retrieval Module for Web Search Results

- Retriever performs representation-based similarity search.
- It retrieves top-k relevant sentences.



# Mock APIs & Mock KG

- Task 2 provides mock APIs to query the provided mock knowledge graph (mock KG).

| | |
|---|---|
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- The mock KG, as a structured knowledge base, offers precise information; however, generating an accurate query is essential for retrieving correct answers.

# Retrieval Module for Mock KG

- The knowledge graph retrieval module follows these three steps:

  a. The LLM generates the query domain and API arguments.

  b. Based on the generated query domain and API arguments, a decision tree is used to sequentially call the appropriate mock APIs.

  c. The results from these mock API calls are provided to the LLM along with retrieved results from web search results.

- We aim to go a step further by leveraging the Model Context Protocol (MCP) to upgrade the knowledge graph retrieval module.

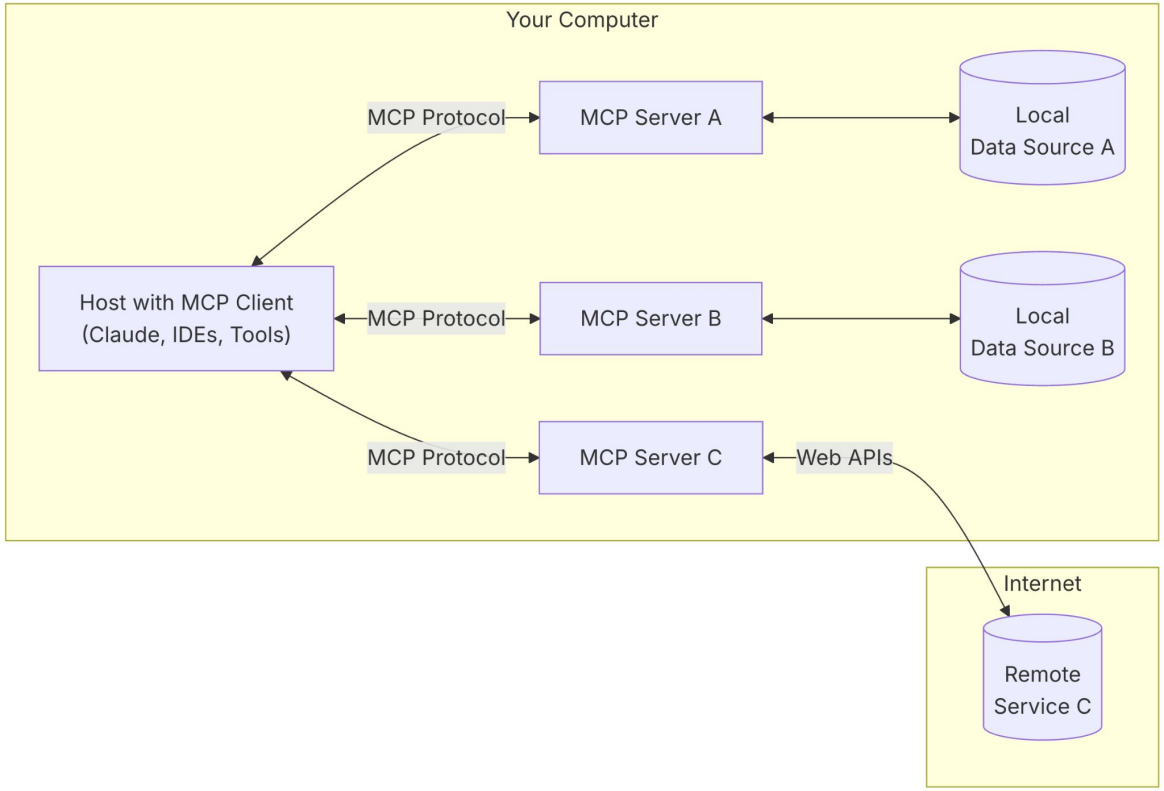
# Model Context Protocol

Model Context Protocol (MCP) is an open protocol that standardizes how applications provide context to LLMs.

- Think of MCP like a USB-C port for AI applications.
- Just as USB-C provides a standardized way to connect your devices to various peripherals and accessories, MCP provides a standardized way to connect AI models to different data sources and tools.

# General Architecture

At its core, MCP follows a client-server architecture where a host application can connect to multiple servers:



# Appendix